ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PRACTICE EXAM 3

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 7 strands: Life and Living (requirements for life, classification, pollination, puberty and pregnancy), Matter and Materials (filtration, pure substances and mixtures, pH, the periodic table), Energy and Change (heat transfer, solar heating, energy transfer, the electricity grid) and Planet Earth and Beyond (Moon phases, seasons, tides, history of astronomy).

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully and answer only what is asked.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which class of vertebrates breathes through gills for its whole life?

A Amphibians B Reptiles C Birds D Fish

1.2 The female part of a flower is called the …

A stamen. B carpel (pistil). C sepal. D petal.

1.3 The main function of the sepals is to …

A attract insects. B make pollen.
C protect the flower while it is a bud. D make seeds.

1.4 Which method separates the different coloured dyes in a sweet's coating?

A Chromatography B Filtration C Decanting D Sieving

1.5 Which of these is a base?

A Vinegar B Baking soda solution
C Lemon juice D Sour milk

1.6 Elements in the same vertical column of the periodic table belong to the same …

A period. B row. C compound. D group.

1.7 Which material is the best conductor of heat?

A Copper B Wood C Plastic D Wool

1.8 The energy stored in food and fuels is …

A kinetic energy. B sound energy.
C chemical potential energy. D light energy.

1.9 Which phase do we see when the Moon is between the Earth and the Sun?

A Full moon B First quarter C New moon D Gibbous moon

1.10 The Earth's axis is tilted at about …

A 23,5° B 10° C 45° D 90°

(10)

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number. If the statement is FALSE, write down the correct word(s) to replace the underlined word(s).

2.1 Amphibians have dry, scaly skin. (2)

2.2 Distillation can be used to separate water from ink. (2)

2.3 Heat from the Sun reaches the Earth by convection. (2)

2.4 A generator changes kinetic energy into electrical energy. (2)

2.5 The Moon shines because it makes its own light. (2)

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3: LIFE AND LIVING

3.1 Name TWO things that all living organisms need from the biosphere to stay alive. (2)

3.2 Name the vertebrate class described in each case: (a) has hair or fur and feeds its young on milk; (b) has dry, scaly skin and lays eggs with leathery shells on land; (c) has moist skin, lays eggs in water and lives both in water and on land; (d) has feathers and a beak. (4)

3.3 Name TWO groups of invertebrates other than arthropods. (2)

3.4 Grasses are pollinated by the wind. (a) Give TWO features of wind-pollinated flowers. (b) Why do wind-pollinated plants produce very large amounts of pollen? (3)

3.5 After fertilisation, what does (a) the ovule and (b) the ovary develop into? (2)

3.6 Name TWO changes during puberty that take place only in girls, and TWO that take place only in boys. (4)

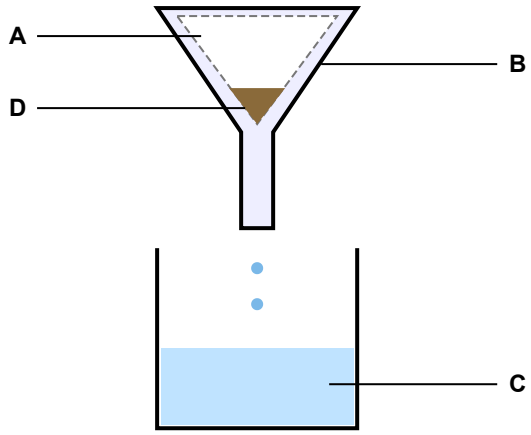
3.7 Explain why a pregnant woman should not smoke or drink alcohol. (2)

3.8 What is a *species*? (1)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 The diagram shows how muddy water is separated.



4.1.1 Name this separation method. (1)

4.1.2 Identify A, B and D. (3)

4.2 Classify each substance as a pure substance or a mixture: (a) air; (b) oxygen; (c) sea water; (d) table salt. (4)

4.3 Universal indicator was used to find the pH of three solutions: P has a pH of 2, Q has a pH of 7 and R has a pH of 12. (a) Which solution is the most acidic? (b) Which solution is neutral? (c) Which solution could be a drain cleaner? (3)

4.4 (a) In what order are the elements arranged in the periodic table? (b) Name TWO elements that are gases at room temperature. (c) Write down the chemical symbol for helium. (4)

4.5 (a) Give TWO ways in which plastic bags harm the environment. (b) Suggest ONE way to reduce the use of plastic bags. (3)

4.6 Name the best method to: (a) remove tea leaves from a cup of tea; (b) get salt from sea water. (2)

[20]

QUESTION 5: ENERGY AND CHANGE

5.1 Thandi makes soup in a pot on a gas stove. Name the way heat is transferred: (a) through the metal bottom of the pot; (b) through the soup; (c) to her face as she stands near the stove. (3)

5.2 Why do some good-quality pots have copper bottoms? (1)

5.3 (a) Why are the panels of a solar water heater painted black? (b) Why is the hot-water tank of a solar water heater insulated? (3)

5.4 A child is on a swing. Describe where on the swing's path the child has the most potential energy and the most kinetic energy, and what happens to the energy between these points. (3)

5.5 An electric kettle uses 2 000 J of electrical energy. 1 800 J heats the water. (a) How much energy is "wasted"? (b) Where does this wasted energy go? (2)

5.6 Classify each energy source as renewable or non-renewable: (a) uranium; (b) biogas; (c) natural gas. (3)

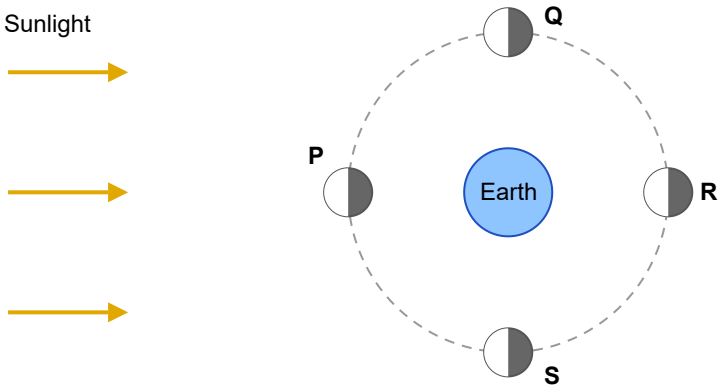
5.7 Arrange these parts of the national electricity grid in the correct order, from where electricity is made to where it is used: step-down transformer, homes, power station, transmission lines, step-up transformer (3)

5.8 (a) Name the company that supplies most of South Africa's electricity. (b) Name ONE energy source other than coal that this company uses. (2)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1 The diagram shows the Moon at four positions (P, Q, R and S) as it orbits the Earth.



The Sun is far to the left of the diagram.

6.1.1 At which position is it (a) new moon and (b) full moon? (2)

6.1.2 Explain why we see a full moon at the position you gave in 6.1.1(b). (2)

6.2 Explain why South Africa is hot in December. (2)

6.3 On about which date is the longest day of the year in South Africa? (1)

6.4 (a) How many high tides are there at a beach in Durban in about one day? (b) The Sun is much bigger than the Moon, yet the Moon has a bigger effect on tides. Why? (2)

6.5 Why must you never look directly at the Sun, even during a solar eclipse? (1)

6.6 Give ONE contribution to astronomy made by each scientist: (a) Ptolemy; (b) Copernicus; (c) Galileo Galilei; (d) Isaac Newton. (4)

6.7 Which direction can be found at night using the Southern Cross? (1)

6.8 (a) What does the MeerKAT radio telescope detect from space? (b) Give TWO reasons why the Karoo was chosen for radio telescopes like MeerKAT and the SKA. (3)

6.9 Why are there fewer hours of daylight in South Africa in winter? (2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER